

PH402 Astrophysics: Final Examination

Write your Name and Roll Number.

The total marks for all the questions together carry more than 50 marks. But your maximum will be restricted to 50 and scaled appropriately.

Your Notes are allowed. No internet or discussion

You can attempt any questions. Present the relevant points, supported by diagrams, expressions or numbers.

Graphs should have Caption, X- and Y- axes marked suitably.

Define the variables used in the expressions.

1. A star of mass 40 solar mass ($M_{\odot}$) has central density $\rho_c = 2000 \text{ kg m}^{-3}$ and central temperature $T_c = 4.5 \times 10^7 \text{ K}$.
It has 6 protons for every α particle in the core, and trace of higher elements.
 - a. Write the **expression** for the **total pressure** at its centre including all contributions. **Define** the terms used in the expression.
 - b. Nuclear energy generation in the core of this star is mainly starting from the CNO cycle reaction ${}^{12}_6\text{C} + {}^1_1\text{H} \rightarrow {}^{13}_7\text{N} + \gamma$,
where C= Carbon nucleus, N= Nitrogen nucleus, H= Hydrogen and γ is a 1.9 MeV γ ray released due to the reaction.
But in a $0.7 M_{\odot}$ Main Sequence Star, the much slower
 ${}^1_1\text{H} + {}^1_1\text{H} \rightarrow {}^2_1\text{D} + e^+ + \nu_e + 0.42 \text{ MeV}$
of PP chain is the first step of the nuclear fusion.
Why is this difference?
 - c. **Which observations** are the direct evidence that above PP chain reaction is operating inside the Sun?
 - d. What new Physics was inferred from above observations? [5 marks]
2. A gas cloud of pure hydrogen molecules is in hydrostatic equilibrium. If it is a sphere of uniform density, **write the relation** between its Mass M , Radius R and Temperature T . [3 marks]
3. A protostar of mass 0.06 solar mass of solar chemical composition collapses to a radius of 0.15 solar radius when its central temperature increases to $3.5 \times 10^6 \text{ K}$.
 - a. Even if it were to collapse to a radius of 0.05 solar radius, its central temperature will not increase further. **State the reason for decrease in core temperature when it collapses further.**
 - b. The protostar emits radiation in infrared wavelengths.
What is the source of energy for this radiation? [3 marks]
4. Evolutionary Track in HR diagram:
 - a. Show the Main Sequence Band, Tip of Red Giant, White Dwarf and Luminosity class Ia in the Hertzsprung–Russell Diagram.

X-axis of the above diagram should have Effective Temperature of the star in the conventional way of higher temperature to the left.
Y-axis should be $\log(\frac{L_*}{L_\odot})$, increasing upwards, where $L_\odot$ is present day solar luminosity.

- b. Show the position of Sun in this diagram.
 - c. **Draw the Evolutionary Track** of an isolated star of $0.9 M_\odot$ of solar abundance from **Zero-Age-Main-Sequence** to its end of evolution.
 - d. Mark the main stages of evolution of this star in the Track shown above. dedik
- [8 marks]

5. White Dwarf Mass-Radius-Central Density:

- a. Write the stellar structure equations for (1) Hydrostatic equilibrium and (2) mass enclosed within a radius r for a spherically symmetric star.
- b. Convert it into a single second order ordinary differential equation in **dimensionless form** after defining new dependent variables $Z = \frac{\rho}{\rho_c}$ and $Y = \frac{P}{P_c}$ where ρ is the density and P is the pressure. Subscript c stands for their central values (P_c, ρ_c).
- c. From this equation, **determine the length scale** for radial distance.
- d. For a Nonrelativistic Degenerate matter, $P \propto \rho^{\frac{5}{3}}$. Hence derive the **Mass-Radius- ρ_c** relation for a White Dwarf well below the Chandrasekhar limit. [11 marks]

6. Rotation curve of Spiral Galaxies:

- a. Draw the circular rotation speed of stars in a galaxy versus radial distance due to a central **sphere of radius $R=1$ kiloparsec (kpc)** having uniform mass density such that the **maximum rotation speed is 180 km/sec**.

The x-axis of your graph should have radial distance from the centre of the galaxy, r , in kpc and y-axis, rotation speed in kilometer per sec.

- b. Draw the circular rotation speed versus radius due to matter of Spherically Symmetric Mass distribution having density profile $\rho(r) \propto \frac{1}{r^2}$, **in the same diagram**.

Rotation speed due to this mass should be 100 km/sec at $r=30$ kpc.

- c. **Draw the observed rotation curve of Milky Way** in the above diagram.
- d. Interpret the above graph (in two sentences). [8 marks]

7. FRW Metric & Standard Model of Cosmology:

- a. State two observations in Cosmology and two philosophical assumptions which lead to standard Friedmann-Robertson-Walker Metric.
- b. **Which Dynamical Equation** of Cosmology and **which observation** imply that Exotic Matter of Negative Pressure dominates the

Universe at present, if we take matter in the fluid approximation with density and pressure.

- c. Derive the expression for the Proper Distance from a source at redshift $z=z_0$ to the observer at $z=0$ for matter obeying the equation of state $P = -\rho$ where P is the pressure and ρ is the energy density, when the spatial curvature scalar $k=0$ and Hubble Constant $=H_0$.

[10 marks]

8. Gravitational Lens:

- a. Einstein Ring is formed due to gravitational lensing of a source at a distance $(D_s) = 20$ kiloparsec by a lens of mass $= 3 \times 10^6$ solar mass, at a distance $D_d = 10$ kiloparsec in front of it.

Einstein Radius of the Ring is 1 arcsecond if the distance from the source to the lens $D_{ds} = 10$ kiloparsec.

Determine the Einstein Radius if the Mass of the Lens is 3×10^{11} solar mass, $D_s = 1500$ Mpc, $D_d = 800$ Mpc, $D_{ds} = 600$ Mpc.

- b. If you find an image of the lensed source at an angular separation of $1''.2$ from the lens, what is its approximate magnification (for the above system)?

- c. If you assume that the lens of the above system is a point mass, what is the **angular separation** between the lens and the lensed source?

[6 marks]

Astronomical & Physical Constants

Parsec		$3.0856 \times 10^{16} \text{ m}$
Mass of the Sun		$1.99 \times 10^{30} \text{ kg}$
Velocity of light in vacuum	c	$3 \times 10^8 \text{ m / sec}$
Planck's constant	h	$6.63 \times 10^{-34} \text{ Joule sec}$
Boltzmann Constant	k	$1.38 \times 10^{-23} \text{ Joule/Kelvin}$
Gravitational Constant	G	$6.67 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ sec}^2$
1 atomic mass unit	a.m.u.	931.49 MeV
Avagadro Number	N	$6.022 \times 10^{23} \text{ per mol.}$
Mass of Hydrogen atom	1.0087 amu	$1.672 \times 10^{-27} \text{ kg}$
Radiation constant	a	$7.6 \text{ J m}^{-3} \text{ K}^{-4}$
Gas Constant	R	$8.3 \text{ J K}^{-1} \text{ mol}^{-1}$